

ShipNoble Index (SNI)

Delivery Division

Comprehensive Technical Specification

Version 2.0 | March 2026 | CEO Reference Document

Classification: Internal / Confidential

Asset Class: Cargo Vans | Unit of Measure: Case Equivalents (CE)

Pricing Effective: Spring 2026

v2.0 Change Summary: 17 of 18 v1.0 gaps resolved. Full pricing matrices integrated. Quality Modifier fully specified. Dynamic SNI defined.

1. Executive Summary

The Delivery Division SNI is a proprietary profitability index that measures the financial performance of ShipNoble's cargo van delivery operations at every level of granularity, from the individual product up to the entire route. Unlike the Transportation Division (which uses palletised freight on straight trucks), the Delivery Division operates with Case Equivalents as its fundamental unit, introduces a physical Capacity SNI layer, and incorporates a Quality Modifier that produces a separate Quality-Adjusted SNI score alongside the financial SNI.

The index answers a single question at every level: Is this unit of work making or losing money, and by how much relative to the 40% target margin?

1.1 Core SNI Formula

$$\text{SNI} = ((R - C) / R) + 0.6$$

Where R = Revenue and C = Total Allocated Cost. The 0.6 offset calibrates the scale:

SNI Value	Margin	Interpretation
1.0	40%	Target margin achieved
0.6	0%	Break-even
< 0.6	Negative	Loss-making
> 1.0	> 40%	Exceeding target

Each 0.1 increment in SNI corresponds to a 10 percentage-point movement in margin.

1.2 Dual Score Architecture

The Delivery Division produces two distinct SNI scores per route:

Financial SNI: The pure profitability score calculated from revenue and cost data. This is the authoritative financial metric and is never modified by quality factors.

Quality-Adjusted SNI: Financial SNI multiplied by the Quality Modifier. This is a separate diagnostic score that reflects both financial performance and operational quality. It can go negative on catastrophic routes. It does not replace or alter the Financial SNI.

$$\text{Quality-Adjusted SNI} = \text{Financial SNI} \times \text{Quality Modifier}$$

1.3 Authoritative Cost Constants

Constant	Value	Derivation / Application
Variable Cost Rate	\$0.39/km	Per-km cargo van operating cost
Labour Rate	\$0.444/min	Driver labour (\$26.64/hr)
Fixed Route Base	\$86.36/route	\$1,900/mo (lease + insurance + parking) / 22 operating days
Van Capacity Baseline	150 CE	Hardcoded maximum for Capacity SNI
Target Profit Margin	40%	SNI = 1.0 at this margin
Residential Surcharge	\$3.50/stop	Applied when residential flag = 1
Large Keg Surcharge	\$10.00/keg	Per 50L or 58L keg

2. Phase 1: Determine the Evaluation State

Before any cost or revenue calculation, the system must determine which temporal state of the SNI is being evaluated.

2.1 Predicted SNI (Pre-Execution)

Calculated before route execution. All inputs derived from the manifest and DMS: predicted time (t_{pred}), predicted distance (d_{pred}), predicted stem costs (SCF_{pred} , SCL_{pred}), and product quantities. Establishes the financial baseline.

2.2 True SNI (Post-Execution)

Calculated after route completion. All inputs derived from the final route status in the DMS: true total time (t_{true}), true distance (d_{true}), true stem costs (SCF_{actual} , SCL_{actual}), actual quantities delivered, and time splits (transit vs. dwell). The DMS system (not ELD hardware) is the authoritative source for all Realised timing data.

The True SNI decomposes labour into transit and dwell components, which the Predicted SNI does not.

2.3 Dynamic SNI (During Loading)

The Dynamic SNI is the live, in-progress version of the Predicted SNI. It recalculates continuously as cases are added to the van during the loading and route-building phase. The routing optimisation team uses the Dynamic SNI as their working metric. Once the route is finalised and dispatched, the Dynamic SNI at that moment becomes the Predicted SNI.

Dynamic SNI = Predicted SNI formula applied to current van contents

The Dynamic SNI uses the same formula as Predicted SNI but with the current (incomplete) loading state rather than the finalised manifest.

2.3.1 Dispatch Threshold

Stop loading when Dynamic SNI reaches 1.0. This corresponds to the 40% target margin. Adding further stops beyond this point risks diluting profitability below target.

2.4 Predicted vs. True Variance Alert

The system triggers an investigation alert when the True SNI falls below 80% of the Predicted SNI value. For example, if Predicted SNI = 0.90, an alert fires if True SNI < 0.72.

Alert Trigger: $\text{True SNI} < (\text{Predicted SNI} \times 0.80)$

3. Phase 2: Capacity SNI (Physical Utilisation)

The Capacity SNI is unique to the Delivery Division. It measures how physically full the cargo van is relative to its hardcoded baseline of 150 Case Equivalents.

3.1 Case Equivalence Matrix

Product Type	Case Equivalent (CE)
12-Wine (12-bottle case)	1.0 CE
6-Wine (6-bottle case)	0.5 CE
Beer Flat	1.0 CE
50L / 58L Keg (large)	1.5 CE
15L, 10L, or 5L Keg (small)	0.5 CE

3.2 Capacity SNI Formula

$$\text{Capacity SNI} = \min(\text{Loaded CEs} / 150, 1.0)$$

Capped at 1.0. A van at or above 150 CE registers as fully utilised.

NOTE: Capacity SNI is a physical metric, not a financial one. It feeds into the Dynamic SNI and dispatch decisions but does not modify the financial SNI formula.

3.3 Minimum Loading Threshold

A Capacity SNI below 0.7 (i.e., fewer than 105 CE loaded) requires management approval before dispatch. Routes below this threshold are structurally at risk of being unprofitable because the fixed route base (\$86.36) is spread across too few revenue-generating cases.

Management Approval Required: Capacity SNI < 0.7 (< 105 CE)

4. Phase 3: Calculate Job Revenue (Ri)

Revenue for each job (stop) is the sum of product-level revenue streams and applicable surcharges.

4.1 Revenue Formula

$$R_i = R_w + R_b + R_k + R_s$$

4.1.1 Wine Revenue (Rw)

$$R_w = Q_w \times ZP_w(\text{zone, quantity})$$

Wine pricing varies by geographic zone and, for Zones 1 and 2, by volume tier. The zone and quantity together determine the per-case rate.

4.1.2 Beer Flat Revenue (Rb)

$$R_b = Q_b \times ZP_b(\text{zone, quantity})$$

Beer flat pricing varies by zone with a volume discount: the first flat is priced higher, with subsequent flats at a reduced rate.

4.1.3 Keg Revenue (Rk)

$$R_k = Q_k \times ZP_k(\text{zone})$$

Keg pricing is a flat per-keg rate by zone with no volume tiers.

4.1.4 Surcharge Revenue (Rs)

$$R_s = (nR \times \$3.50) + (nLK \times \$10.00)$$

nR = Residential flag (1 if residential, 0 if commercial). **nLK** = Number of large kegs (50L or 58L) on the job.

4.1.5 Ancillary Revenue Streams

The following ancillary charges generate additional revenue per job where applicable:

Service	Rate	Applies To
Trillium Pickup (daily weekday)	\$2.00/unit	Wine cases and beer flats
Wine Pick-up from Location	\$5.00/case	Wine (\$10 minimum)
Flat Pick-up from Location	\$3.00/flat	Beer flats (\$10 minimum)
Keg Pick-up from Location	\$10.00/keg	Kegs (\$10 minimum)
Empty Keg Pickup	Included	Every keg delivery

NOTE: Fuel surcharge is absorbed by ShipNoble up to \$1.40/L. No additional handling fees are charged. These are all-in prices.

4.2 Zone Pricing Matrices (Spring 2026)

The Delivery Division uses a three-zone structure for kegs and beer flats. Wine pricing uses the same three macro-zones but with finer geographic granularity and volume-based tiers within Zones 1 and 2.

4.2.1 Zone Definitions

Zone	Label	Geography
1	Toronto, Mississauga & GTA	Toronto, Mississauga, and Greater Toronto Area
2	Major Cities	Vaughan, Nobleton, King City, Woodbridge, Ajax, Pickering, Cobourg, Whitby, Hamilton, Oakville, Burlington, Brampton, Aurora, Newmarket, Markham, Richmond Hill, Thornhill, Stouffville, Niagara Falls, Grimsby, St. Catharines, Vineland, Beamsville, Crystal Beach, Orangeville, Georgetown, Milton, Bolton, Caledon, Alton, Mono, Prince Edward County, Kingston, Belleville, Port Hope, London, Guelph, Cambridge, Elora, Kitchener, Stratford, Waterloo, Peterborough, Uxbridge, Lindsay, Port Perry, Millbrook, Barrie, Orillia, Thornbury, Collingwood, Innisfil, Ottawa, Stittsville, Kanata, Manotick, Carleton Place, Nepean
3	Rural Cities	Windsor, Chatham, Amherstburg, Muskoka, Cornwall, Hawkesbury, Russell, Greely, Rockland, Cumberland, Prescott, Merrickville, Richmond, Arnprior, Renfrew, Pembroke, Petawawa, Calabogie, Listowel, Grand Bend, Sudbury, North Bay, Sarnia

4.2.2 Keg Pricing Matrix

Zone	Geography	Rate
Zone 1	Toronto, Mississauga & GTA	\$21.00 per keg
Zone 2	Major Cities	\$26.00 per keg
Zone 3	Rural Cities	\$30.00 per keg

Additional: \$10.00 surcharge per 50L/58L keg. Empty keg pickup included with every keg delivery.

4.2.3 Beer Flat Pricing Matrix

Zone	Geography	1 Flat	2+ Flats (each)
Zone 1	Toronto, Mississauga & GTA	\$12.00	\$6.00
Zone 2	Major Cities	\$16.00	\$8.00
Zone 3	Rural Cities	\$18.00	\$9.00

4.2.4 Wine Case Pricing Matrix

Wine pricing uses geographic sub-tiers within the three macro-zones, with volume discounting in Zones 1 and 2.

Zone 1: Toronto & Mississauga

Volume Tier	Per Case Rate
1 case	\$13.00
2-3 cases	\$10.00
4-5 cases	\$9.50
6-8 cases	\$8.50
9-10 cases	\$7.75
11+ cases	\$7.25

Zone 2: Major Cities (sub-tiered)

Sub-Zone Geography	Per Case Rate
Vaughan, Nobleton, King City, Woodbridge, Ajax, Pickering, Cobourg, Whitby, Hamilton, Oakville, Burlington, Brampton, Aurora, Newmarket, Markham, Richmond Hill, Thornhill, Stouffville	1 case: \$15.00 / 2+: \$12.00
Niagara Falls, Grimsby, St. Catharines, Vineland, Beamsville, Crystal Beach, Orangeville, Georgetown, Milton, Bolton, Caledon, Alton, Mono	\$15.50 flat
Prince Edward County, Kingston, Belleville, Port Hope, London, Guelph, Cambridge, Elora, Kitchener, Stratford, Waterloo	\$16.00 flat
Peterborough, Uxbridge, Lindsay, Port Perry, Millbrook, Barrie, Orillia, Thornbury, Collingwood, Innisfil	\$16.50 flat
Ottawa, Stittsville, Kanata, Manotick, Carleton Place, Nepean	\$17.00 flat

Zone 3: Rural Cities

Sub-Zone Geography	Per Case Rate
Windsor, Chatham, Amherstburg, Muskoka	\$18.00 flat
Cornwall, Hawkesbury, Russell, Greely, Rockland, Cumberland, Prescott, Merrickville, Richmond, Arnprior, Renfrew, Pembroke, Petawawa, Calabogie, Listowel, Grand Bend	\$20.00 flat
Sudbury, North Bay, Sarnia	\$23.00 flat

4.3 Predicted vs. True Revenue

For Predicted SNI, revenue derives from manifest quantities and zone assignment at dispatch. For True SNI, revenue derives from final route status, accounting for partial deliveries, refusals, and quantity adjustments. If a job is entirely refused, True Revenue = \$0 but costs remain allocated.

5. Phase 4: Calculate Total Job Costs (TCi)

Each job receives an allocated share of five cost categories.

5.1 Total Cost Formula

$$TC_i = V_i + LC_i + F_i + SOH_i + E_i$$

5.2 Variable Cost (Vi)

$$V_i = d_i \times \$0.39/\text{km}$$

d_i = distance in km for the job (d_{pred} or d_{true} depending on evaluation state).

5.3 Labour Cost (LCi)

5.3.1 Predicted State

$$LC_i(\text{pred}) = t_{\text{pred}} \times \$0.444/\text{min}$$

Single undifferentiated time allocation covering transit and dwell combined.

5.3.2 True State

$$LC_{\text{transit}} = t_{\text{transit_true}} \times \$0.444/\text{min}$$

$$LC_{\text{drop}} = t_{\text{drop_true}} \times \$0.444/\text{min}$$

$$LC_i(\text{true}) = LC_{\text{transit}} + LC_{\text{drop}}$$

$t_{\text{transit_true}}$ = time from previous stop to arrival. $t_{\text{drop_true}}$ = time parked at the stop. This decomposition isolates transit efficiency from dwell efficiency.

5.4 Fixed Allocation (Fi)

$$F_i = (t_i / T_{\text{route}}) \times \$86.36$$

The \$86.36 fixed route base (\$1,900/month divided by 22 operating days; includes lease, insurance, and parking) is allocated to each job proportionally by time. A job consuming more route time bears a proportionally larger share.

5.5 Stem Overhead (SOHi)

$$SOH_i = (SCF + SCL) / n_{\text{ts}}$$

SCF = First Job Stem Cost. **SCL** = Last Job Stem Cost. **n_{ts}** = Total number of stops on the route.

Stem costs are split equally across all stops. This is intentional for the Delivery Division (versus time-proportional in Transportation), reflecting the higher stop count and lower per-stop duration variance in cargo van routes.

5.5.1 Stem Cost Composition

Stem costs are variable-only. They do not include any portion of the fixed route base.

$$SCF = (\text{stem_distance} \times \$0.39/\text{km}) + (\text{stem_time} \times \$0.444/\text{min})$$

$$SCL = (\text{return_distance} \times \$0.39/\text{km}) + (\text{return_time} \times \$0.444/\text{min})$$

5.6 Error Cost (Ei)

E_i = Sum of all Cost of Errors (True SNI only)

Error costs are applied only in the True SNI evaluation state. They attach to the specific stop where the error occurred, not spread across the route. Error costs are additive to TC_i, reducing the job's SNI.

OPEN ITEM: OPEN ITEM: Error Cost Dollar Values

The specific dollar amounts per error category (breakage, mis-delivery, returns processing, etc.) are pending definition by Operations. This is the sole remaining open item in the specification. The allocation rule (stop-specific) is confirmed.

6. Phase 5: Job Profit and Route Financials

6.1 Job Profit

$$P_i = R_i - TC_i$$

6.2 Job SNI

$$\text{Job SNI} = ((R_i - TC_i) / R_i) + 0.6$$

6.3 Route Aggregation

$$\text{Route Revenue} = \text{Sum}(R_i) \text{ for } i = 1 \text{ to } n$$

$$\text{Route Total Cost} = \text{Sum}(TC_i) \text{ for } i = 1 \text{ to } n$$

$$\text{Route SNI} = ((\text{Route Revenue} - \text{Route Total Cost}) / \text{Route Revenue}) + 0.6$$

6.4 Route Target Profit

$$\text{Route Target Profit} = \text{Route Revenue} \times 0.40$$

A Route SNI of 1.0 means the route achieved exactly the 40% target margin.

6.5 Static SNI

$$\text{Static SNI} = (\text{Max Revenue} - \text{Ideal Total Costs}) / \text{Target Profit}$$

The Static SNI represents the theoretical maximum profitability if the route were executed perfectly at full capacity (150 CE). Max Revenue assumes full loading at zone rates; Ideal Total Costs assumes zero delays, deviations, or errors. The Static SNI serves purely as a ceiling benchmark.

NOTE: The Static SNI is nearly impossible to achieve in practice. It exists to contextualise Predicted and True SNI values as a percentage of theoretical potential.

7. Phase 6: Quality Modifier (QM)

The Quality Modifier produces the Quality-Adjusted SNI, a separate diagnostic score that sits alongside the financial SNI. It does not replace, override, or alter the financial SNI.

7.1 Quality Modifier Formula

$$QM = 1.0 - \text{Sum(Penalties)} + \text{Sum(Rewards)}$$

Baseline of 1.0. Penalties reduce; rewards increase. No floor is applied. The QM can go negative on catastrophic routes, which is intentional: a negative Quality-Adjusted SNI is a clear signal of severe operational failure.

7.2 Application

$$\text{Quality-Adjusted SNI} = \text{Financial SNI} \times QM$$

The Quality-Adjusted SNI is a separate metric. It appears alongside the Financial SNI in all reporting but never substitutes for it.

7.3 Penalty Schedule

Event	Penalty	Trigger Condition
Collision	-0.05	Any collision involving the van, regardless of fault
Breakage	-0.25	Any breakage of product after leaving the warehouse
Ticket	-0.05	Any traffic or parking ticket issued to the driver
Incomplete Job	-0.25	Each incomplete job (partial or refused delivery)
1-Star Review	-2.00	Merchant rates delivery 1 out of 5 stars
2-Star Review	-1.50	Merchant rates delivery 2 out of 5 stars
3-Star Review	-1.00	Merchant rates delivery 3 out of 5 stars
4-Star Review	-0.50	Merchant rates delivery 4 out of 5 stars
5-Star Review	0.00	No penalty (neutral)

Review penalties are linear: each star below 5 adds -0.50 to the penalty. A 1-star review is the single most punitive event in the system, exceeding all other penalties combined.

7.4 Reward Schedule

Event	Reward	Trigger Condition
Pristine Run	+0.05	Zero penalties triggered on the entire route
5-Star Review	+0.05	Merchant rates delivery 5 out of 5 stars

NOTE: A 5-star review is both a neutral penalty (0.00) and a positive reward (+0.05). Net effect of a single 5-star review on a pristine route: $QM = 1.0 + 0.05 \text{ (pristine)} + 0.05 \text{ (5-star)} = 1.10$.

7.5 Worked Example: Catastrophic Route

Scenario: collision, breakage, 2 incomplete jobs, and a 1-star review.

$$QM = 1.0 - 0.05 - 0.25 - 0.25 - 0.25 - 2.00 = -1.80$$

If Financial SNI = 0.85, then Quality-Adjusted SNI = $0.85 \times (-1.80) = -1.53$. The negative score is intentional and signals catastrophic operational failure requiring immediate investigation.

7.6 Worked Example: Excellent Route

Scenario: no issues, 5-star review.

$$QM = 1.0 + 0.05 + 0.05 = 1.10$$

If Financial SNI = 0.95, then Quality-Adjusted SNI = $0.95 \times 1.10 = 1.045$. The route exceeded both financial and quality targets.

8. SNI Hierarchy

The Delivery Division SNI operates at multiple levels of aggregation, each serving a different analytical purpose.

8.1 Hierarchy Overview

Level	Definition	Primary Audience	Status
Static SNI	Theoretical max at full capacity	Executive	Fully specified
Dynamic SNI	Live recalculation during loading	Routing Team	Fully specified
Predicted SNI	Finalised route, pre-execution	Operations	Fully specified
True SNI	Actual execution results	Operations	Fully specified
Route SNI	Aggregated route P&L	Operations	Fully specified
Job SNI	Individual stop profitability	Operations	Fully specified
Client SNI	Rolling average per merchant	Commercial	Fully specified
Quality-Adj. SNI	Financial SNI x QM	Operations / HR	Fully specified
Capacity SNI	Physical utilisation metric	Routing Team	Fully specified
Product SNI	Per-case unit economics	Finance	Basic (v2 detail)
Loading SNI	Warehouse-to-van cost	Warehouse Ops	Deferred to v2

8.2 Client SNI

$$\text{Client SNI} = \text{Sum(All Job SNIs for Client)} / \text{Total \# of Jobs for Client}$$

Rolling average across all jobs for a specific merchant. Answers: 'Is this client sustainably profitable?'

8.3 Product SNI (Basic Diagnostic)

$$\text{Cost Per CE} = \text{Total Job Cost} / \text{Total CEs on Job}$$

The Product SNI provides a basic unit economics diagnostic at the case-equivalent level. It answers: 'What is the average cost to deliver one CE on this job?'

Detailed per-case cost attribution (pick/pack/wrap time allocation, space factors, handling penalties, and warehouse-level cost decomposition) will be fully specified under the Loading SNI module in v2. For v1, the Cost Per CE metric provides directional visibility into unit economics.

DEFERRED TO v2: Detailed Product SNI cost attribution methodology (pick/pack/wrap, space factors, handling penalties) will be specified under Loading SNI in v2.

8.4 Driver Efficiency (ROE)

$$\text{ROE} = \text{Predicted Cost (Speed)} / \text{True Cost (TTrue)}$$

Compares predicted cost to actual cost across a driver's collection of routes. ROE > 1.0 = driver executes below predicted cost. ROE < 1.0 = driver is more expensive than planned. Should be evaluated over a statistically significant sample to avoid penalising for one-off external factors.

9. Architectural Distinctions: Delivery vs. Transportation

Dimension	Delivery Division	Transportation Division
Asset Class	Cargo vans	Straight trucks / Isuzu
Unit of Measure	Case Equivalents (CE)	Pallets
Capacity SNI	Yes (capped 1.0, 150 CE)	No (removed)
Variable Rate	\$0.39/km	\$0.55/km
Labour Rate	\$0.444/min (\$26.64/hr)	\$0.7462/min (\$44.77/hr)
Fixed Base	\$86.36/route (\$1,900/22d)	\$156.59/day (\$3,445/22d)
Stem Allocation	Equal-split across stops	Time-proportional
Stem Composition	Variable-only	Variable-only
Quality Modifier	Yes (dual score output)	No
Revenue Model	3 zones, product-type rates, volume tiers	45 zones, pallet + \$75 add'l
Dispatch Threshold	Dynamic SNI = 1.0 (stop loading)	N/A
Min Loading	Cap SNI < 0.7 = mgmt approval	N/A
Dock SNI	Not applicable	Yes (time-based)

10. Open Items and Deferred Modules

10.1 Remaining Open Item (1 of 18)

#	Description	Severity	Owner	Notes
G8	Error cost dollar values per error type	Medium	Operations	Allocation rule (stop-specific) confirmed. Dollar values pending.

10.2 Deferred Modules (v2 Roadmap)

Module	Scope	Dependency
Loading SNI	Warehouse-to-van cost attribution: pick/pack/wrap time, space factors, handling penalties, per-case cost decomposition	Warehouse ops data
Detailed Product SNI	Full per-case profitability with attributed warehouse costs feeding from Loading SNI	Loading SNI
Handoff SNI	End-to-end merchant cost: order receipt through van departure, including pre-delivery processing costs	Loading SNI + CRM data
Inter-Branch Transfer SNI	Profitability of product transfers between ShipNoble branches/warehouses	Branch costing model

10.3 v1.0 to v2.0 Resolution Summary

Of the 18 gaps identified in v1.0, 17 have been resolved in this v2.0 specification:

Resolved by direct specification: G1 (fixed base derivation), G2 (variance threshold), G3 (min loading), G4/G5 (zone pricing), G6 (keg surcharge scope), G7 (stem composition), G9 (QM application formula), G10 (review scale), G11 (QM floor decision), G12 (review source), G17 (Dynamic SNI), G18 (dispatch threshold).

Resolved by scoping decision: G13 (Handoff SNI deferred), G14 (Inter-branch deferred), G15 (Loading SNI deferred), G16 (Product SNI simplified for v1).

Remaining open: G8 (error cost dollar values, owner: Operations).

11. Next Steps

1. Resolve G8 (Error Cost Dollar Values): Operations to define the specific categories and dollar amounts for each error type. Allocation rule (stop-specific) is confirmed.

2. System Implementation: This specification is implementation-ready for all components except error cost pricing. DMS integration can proceed for all six phases.

3. Loading SNI (v2): Begin scoping the warehouse-to-van cost attribution model as a separate workstream. This will unlock the detailed Product SNI and Handoff SNI modules.

4. Pricing Review Cadence: The Spring 2026 pricing matrices are now embedded in this specification. Establish a review cadence (quarterly or seasonal) to update zone rates and ensure the SNI calculations reflect current pricing.

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